Logic & Philosophy

A Complete Foundations Study Guide

- What Logic Is — definitions, aims, value & types
- The Laws of Thought
- The Nature of Arguments — premises, conclusions & validity
- Identifying & Analysing Arguments
- Definition — types, purposes & rules

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How to use this guide

Sections build on one another: start with what logic *is*, learn the laws that govern correct thought, then study arguments — their nature, how to spot them, and finally how to define the terms they use precisely. **Key terms** are highlighted, **logical forms** are flagged in gold, and worked examples show every premise and conclusion in full.

1 What Is Logic?

IN ONE LINE

Logic is the branch of philosophy that studies the principles and methods for telling **good (correct) reasoning** from **bad (incorrect) reasoning** — that is, the study of how we evaluate arguments.

1.1 Defining logic

There are many definitions of logic offered by different authors, but they all share one common emphasis: logic deals with **thinking** and the **laws that guide it**. Rather than memorising one phrasing, it helps to read several and draw out what they have in common.

Scholar	Definition of logic
A. G. A. Bello	The study of the principles and techniques of distinguishing good arguments from bad arguments.
I. M. Copi & Carl Cohen	The study of the methods and principles used to distinguish correct reasoning from incorrect reasoning.
Nolt, Rohatyn & Varzi	The study of argument.
Patrick J. Hurley	The organised body of knowledge, or science, that evaluates arguments.

More precisely, logic is the study of methods for evaluating *whether the premises of an argument adequately support — or provide good evidence for — its conclusion*.

ESSENTIAL LESSONS FROM THE DEFINITIONS

1. Logic is a **study** (an organised, systematic body of knowledge).
2. It studies **arguments** — specifically, how to determine whether an argument is good or bad.

Working definition

Logic is a **branch of philosophy** — and also a **tool of philosophy** — in which we learn the principles for distinguishing good arguments from bad arguments.

1.2 The aim of logic

Logic aims at achieving three things:

1. To give us **criteria** we can use to test whether an argument is good or bad, valid or invalid, sound or unsound, deductive or inductive.
2. To develop a **system of methods and principles** we can use as criteria for evaluating the arguments made by others.
3. To develop methods, principles and techniques that serve as **guides for constructing arguments of our own**.

1.3 Why logic matters — in philosophy and beyond

Much of what philosophers do involves evaluating other people's arguments and constructing arguments of their own. Logic lets a philosopher assess others' reasoning while building arguments that are not *fallacious* — that is, not filled with errors. People who never study logic can still reason, but the study of logic lets one analyse arguments with greater **precision**, **system** and **confidence**.

> **IN THEIR WORDS**
> "Through the study of logic, one learns strategies for thinking well, common errors in reasoning to avoid, and effective techniques for evaluating arguments." — **C. Stephen Layman**

Logic is not only important to philosophy. It is also of great importance to **mathematics**, **linguistics**, **semantics**, **engineering** and **computer science**.

1.4 The value of logic

> "All our dignity lies in thought." — **Blaise Pascal**

A training in logic develops abilities and dispositions that go well beyond the classroom:

- It enhances our capacity to **construct good arguments**.
- It enables **critical and reflective analysis** — separating the essentials from the inessentials.
- It helps us **think clearly**.
- It shapes **character and attitude**: a trained mind is rational, intelligently alert, and slow to accept others' ideas without proper scrutiny.
- It makes one more likely to **question one's own prejudices** and rationalisations.
- It helps in **detecting fallacies** in argument.
- It is useful in the **law court**.

1.5 Types of logic

Type	What it studies
Informal logic	Arguments **without form**, in natural (everyday) language; develops tools for treating ordinary discourse. The study of informal fallacies is a good example.
Formal logic	Arguments **with form** — the study of logic by its formal content. E.g. Modus Ponens, Modus Tollens, Hypothetical Syllogism.
Symbolic logic	The study of the symbols used to denote propositions, terms and relations, in order to assist reasoning.
Propositional logic	The logical relationships and properties of **propositions**.
Predicate logic	Logic that analyses the internal structure of propositions (subjects and predicates).

Type	What it studies
Inductive logic	The principles and techniques for handling **inductive arguments** — arguments whose premises do not give conclusive support to the conclusion.
Mathematical logic	An extension of symbolic logic into areas such as model theory, proof theory and set theory.

2 The Laws of Thought

The **Laws of Thought** are the basic principles of correct reasoning. There are three of them. Each can be written in a compact logical form, where **p** stands for any statement and **~** means "not".

Law	Plain statement	Logical form
Identity	If a statement is true, then it is true; every statement is identical with itself.	p ⊃ p
(Non-)Contradiction	No statement can be both true and false at the same time.	p · ~p
Excluded Middle	Every statement is either true or false — there is no middle ground.	p ∨ ~p

2.1 Law of Identity

If any statement is true, then it is true; in other words, every single statement is identical with itself. In logical form: **p ⊃ p**.

EXAMPLE

If the statement "It is raining" is true, then it is true.

2.2 Law of (Non-)Contradiction

No statement can be both true and false at the same time. Every statement of the form **p · ~p** is self-contradictory and therefore false.

EXAMPLE

No one can hold that "Buhari is the President of Nigeria" is true and at the same time hold that "Buhari is *not* the President of Nigeria" is true. To do so is to contradict oneself — the statement cannot be true and false at the same time.

2.3 Law of Excluded Middle

A statement is either true or false. Every statement must be one of the two; there is no middle ground. If a statement is not true, then it is false. In logical form: **p ∨ ~p**.

EXAMPLE

The statement "It is raining" is either true or false. There is no third option.

3 The Nature of Arguments

The main concern of logic is **arguments**: learning the principles by which we can advance, assess and evaluate them.

3.1 Why study arguments?

- **In philosophy:** all of philosophy may be described as presenting arguments and evaluating other people's arguments — so argument plays a critical role.
- **In daily life:** whether we realise it or not, people constantly present arguments to win us over to their side.
- **In the academic disciplines:** intellectual activity is about giving good justification for the views we hold, and evaluating the arguments made by others in our field.

> **AN ANALOGY**
> "Arguments serve the same purpose for philosophers as proofs serve the mathematician, or experimental demonstration serves the scientist, or archaeological evidence and archival documents serve the historian." — **Peter O. Bodunrin**

3.2 What is an argument?

There is a difference between the everyday and the philosophical use of the word "argument."

Usage	Meaning
To a layman	A verbal dispute that may result in a quarrel or disagreement.
To a philosopher	A set of declarative sentences divided into two groups — the **premise(s)** and the **conclusion** — where the premises give support for accepting the conclusion.

> **DEFINITION — ARGUMENT**
> An **argument** is a set of declarative sentences (propositions) with two parts — a **conclusion** and one or more **premises** — in which the premises give support for the acceptance of the conclusion.

How scholars define it

Scholar	Definition
A. G. A. Bello	An argument is a type of inference — a claim to a new piece of information on the basis of something else we already know. Any group of propositions, one of which is claimed (rightly or wrongly) to follow from the others, which are regarded as providing evidence for its truth.
C. S. Layman	A set of statements, one of which (the conclusion) is affirmed on the basis of the others (the premises).
Francis Offor	A group of statements in which one part, the conclusion, is claimed to follow from the others, the premises.

3.3 The building block: the proposition

The basic constituent of an argument is the **proposition** — a declarative sentence that can be either true or false. A proposition therefore has a **truth value**. Note carefully: individual propositions are true or false, but *arguments* themselves are never called true or false.

EXAMPLES OF PROPOSITIONS

1. It is raining. 2. Nigeria is a black nation. 3. Omolara is a girl. 4. Charles is hardworking.

3.4 The structure of an argument

An argument has a structure defined by two parts and the relationship between them:

- It has a **conclusion** and one or more **premises**.
- There must be a **relationship** between them: the premises should provide sufficient ground for the conclusion.
- If the premises provide *no* ground for accepting the conclusion, then there is **no argument**.
- The propositions must be connected and intelligible — without connection between the propositions, there is no argument.

RELATIONSHIP MATTERS — COMPARE

No real argument: "Black is beautiful. Kunle is a fine man. Therefore the car is very fast." — the propositions are unrelated.
A genuine argument: "All men are mortal. Socrates is a man. Therefore Socrates is mortal." — the premises support the conclusion.

Conclusion vs. premise

Part	Role in the argument
Conclusion	The claim one expects others to accept; the proposition that relies on other propositions for its justification.
Premise	The body of evidence given in support of the claim — the reason offered for accepting the conclusion; it provides the ground for the conclusion.

The support premises give a conclusion may be **weak**, **moderate** or **strong**.

3.5 Good arguments vs. bad arguments

Category	Description
Good argument	An argument in which the premises **really do support** the conclusion.
Bad argument	An argument in which the premises **do not support** the conclusion, even though they are claimed to.

3.6 Types of argument

Arguments are classified along several lines: formal / informal, valid / invalid, sound, and deductive / inductive.

Formal arguments

Formal arguments are arguments **with form** — they must take a particular shape. Two classic forms:

Form	Pattern	Worked example
Modus Ponens	$p \supset q$ p $\therefore q$	If it rains then the ground is wet. **P1:** If it rains then the ground is wet. **P2:** It rains. **C:** The ground is wet.
Modus Tollens	$p \supset q$ $\sim q$ $\therefore \sim p$	**P1:** If it rains then the ground is wet. **P2:** The ground is not wet. **C:** It did not rain.

Informal arguments

Informal arguments are arguments **without form**, studied in informal logic. They appear in ordinary language: everyday discussion, advertising, political debate, legal argument, and the social commentary of newspapers, television and the web. Informal fallacies are a good source of examples.

EXAMPLE — INFORMAL ARGUMENT
"Buy Omo Detergent, because 99% of Nigerians use this detergent."

Valid, invalid and sound arguments

Term	Definition
Valid argument	One in which, **if the premises are true, the conclusion cannot be false**.
Invalid argument	One in which the premises may be true yet the conclusion is false.
Sound argument	A **valid** argument that **also has all true propositions** (true premises and a true conclusion).

VALID & SOUND — THE SOCRATES EXAMPLE
P1: All men are mortal — TRUE **P2:** Socrates is a man — TRUE **C:** Socrates is mortal — TRUE.
This argument is valid (true premises force a true conclusion) and, because every proposition is true, it is also **sound**.
Invalid version: same true premises but concluding "Socrates is *not* mortal" — the conclusion is false, so the argument is invalid.

Deductive vs. inductive arguments

Type	How premises support the conclusion	Example
Deductive	Premises give **full** support: if you accept the premises you cannot reject the conclusion — it necessarily follows.	All Nigerians are blacks. All blacks are hardworking. Therefore Kunle, who is a Nigerian, is black and hardworking.

Type	How premises support the conclusion	Example
Inductive	Premises give **partial** support: you may accept the premises yet reject the conclusion — it does not necessarily follow.	Abacha (military head of state) was a dictator. Babangida (military head of state) was a dictator. Therefore all military heads of state are dictators.

4 Identifying & Analysing Arguments

OBJECTIVES

Learn to (i) **identify** arguments, (ii) **analyse** them into premises and conclusion, and (iii) distinguish **conclusion indicators** from **premise indicators**.

4.1 Two ways to identify an argument

1. Through the presence of **indicators** (signal words).
2. Through asking ourselves **two important questions**.

4.2 Indicators

Indicators are signal words whose presence tells us we are dealing with an argument, and which part is the premise and which is the conclusion. There are two kinds.

Conclusion indicators	Premise indicators
therefore, wherefore, thus, consequently, we may infer, accordingly, we may conclude, so, hence, it follows that, it implies that	since, because, for, as, follows from, as shown by, for the reason that, inasmuch as, as indicated by, in view of the fact that

REFERENCE

See *Essentials of Logic*, p. 15 (conclusion indicators) and p. 16 (premise indicators).

4.3 The two-question test

Question	What it identifies
What position are we asked to accept?	The part that answers this is the **conclusion**.
Why are we asked to accept this position?	The part that answers this is the **premise**.

4.4 Analysing an argument

To analyse an argument, separate the conclusion from the premise(s). Write out the **premises first**, then state the **conclusion**.

Worked example 1 — indicator words

ARGUMENT

“All physicians are university graduates, **so** all members of the Nigerian Medical Association must be university graduates, **since** all members of the Nigerian Medical Association are physicians.”

Part	Statement
Premise 1	All physicians are university graduates.

Part	Statement
Premise 2	All members of the Nigerian Medical Association are physicians.
Conclusion	All members of the Nigerian Medical Association must be university graduates.

Worked example 2

ARGUMENT

"No one was present when life first appeared on earth. **Therefore**, any statement about life's origin should be considered as theory, not fact."

Part	Statement
Premise 1	No one was present when life first appeared on earth.
Conclusion	Any statement about life's origin should be considered as theory, not fact.

Worked example 3 — indicator in the middle

ARGUMENT

"The turkey vulture is called that name **because** its red featherless head resembles the head of a wild turkey."

Part	Statement
Premise 1	Its red featherless head resembles the head of a wild turkey.
Conclusion	The turkey vulture is called that name.

Worked example 4 — using the two questions

ARGUMENT

"The Food and Drug Management should stop all cigarette sales immediately. We know one thing, at least: that cigarette smoking is the leading preventable cause of death."

WHAT are we asked to accept? → The Food and Drug Management should stop all cigarette sales immediately. (*conclusion*)

WHY? → Because cigarette smoking is the leading preventable cause of death. (*premise*)

Worked example 5

ARGUMENT

"**Since** the elderly have always had a higher cancer rate, **and since** we now have more older citizens, the absolute increase in the number of cancer deaths is not an indication of any kind of 'environmental breakdown'."

Part	Statement
Premise 1	The elderly have always had a higher cancer rate.
Premise 2	We now have more older citizens.
Conclusion	The absolute increase in the number of cancer deaths is not an indication of any kind of 'environmental breakdown'.

Worked example 6

ARGUMENT

"**Since** witch-doctors are not supposed to serve malevolent ends, **and since** they often use medicine or magic to drive off or destroy witches, they are best classified as good magicians or wizards."

Part	Statement
Premise 1	Witch-doctors are not supposed to serve malevolent ends.
Premise 2	They often use medicine or magic to help drive off, or destroy, witches.
Conclusion	They are best classified as good magicians or wizards.

5 Definition

We often need to look up the meaning of a new word, or to provide definitions of concepts we are using. Knowing the **types**, **purposes** and **rules** of definition makes us better at both tasks.

5.1 Two key technical terms

Term	Meaning
Definiendum	The concept being given meaning — i.e. the word that one is defining.
Definiens	The group of words used to provide the meaning of the word being defined.

5.2 Six types of definition

Type	What it does
Stipulative	Used when a concept appears for the **first time**; the user stipulates how it should be employed. Both definiendum and definiens are new and set by the person introducing them.
Lexical	Reports the **established**, conventional meaning of a word already in use (as a dictionary does).
Precising	Reduces the **vagueness** of a borderline term by stating precisely what one means — e.g. fixing exactly who counts as "youth."
Theoretical	Aims at a **theoretically adequate** account of a disputed concept within a discipline — e.g. experts debating "democracy."
Persuasive	Aims to **persuade** — to influence attitudes or evoke emotions about the definiendum.
Ostensive	Defines by **pointing** to the object being defined.

5.3 Purposes of definition

- To **increase vocabulary** — learn the meaning and usage of a new word (as when we check a dictionary).
- To **eliminate ambiguity** — clarify which sense of a multi-meaning word is intended in a context.
- To **reduce vagueness** — state clearly the limit of a concept whose borderline is unclear.
- To **explain theoretically** — definitions given by professionals within a field of study.
- To **influence attitudes** — evoke positive or negative feelings about the definiendum.
- To **resolve differences** — settle disputes that arise only because parties use a term in different senses.

5.4 Five rules of definition

Five rules have been developed to guide the giving of good definitions:

Rule	Explanation
1. State the essential attributes of the species	Name a characteristic that distinguishes the thing from others, rather than an attribute it merely shares with them.
2. Must not be circular	The definiendum must not appear in the definiens — do not use the word to define itself.
3. Neither too broad nor too narrow	Too broad: it captures things it should not. Too narrow: it excludes things it should include.
4. Avoid ambiguous, obscure or figurative language	The purpose of a definition is clarity; unclear language only adds confusion.
5. Should not be negative where it can be affirmative	Tell us what a concept *is*, not merely what it is not — there are countless things any term does not mean.

★ Quick-Revision Summary

Use this one-page recap to test yourself before an exam. Cover the right column and try to recall each point.

Topic	Key point to remember
What logic is	The study of methods/principles for distinguishing correct reasoning from incorrect reasoning — i.e. evaluating arguments.
Aim of logic	Provide criteria to test arguments; build methods to evaluate others' arguments; guide construction of our own.
Law of Identity	$p \supset p$ — a true statement is true; every statement is identical with itself.
Law of Non-Contradiction	$p \cdot \sim p$ is always false — nothing is both true and false at once.
Law of Excluded Middle	$p \vee \sim p$ — every statement is either true or false; no middle ground.
Argument	A set of propositions: premise(s) offering support for a conclusion.
Proposition	A declarative sentence with a truth value (true or false); arguments themselves are never true/false.
Good vs. bad	Good = premises really support the conclusion; bad = they don't, though claimed to.
Valid / Sound	Valid = true premises can't yield a false conclusion. Sound = valid + all propositions true.
Deductive / Inductive	Deductive = full support (conclusion necessarily follows). Inductive = partial support.
Conclusion indicators	therefore, thus, hence, so, consequently, it follows that …
Premise indicators	since, because, for, as, in view of the fact that …
Two-question test	“What are we asked to accept?” = conclusion. “Why?” = premise.
Definiendum / Definiens	The term being defined / the words doing the defining.
Six types of definition	Stipulative, lexical, precising, theoretical, persuasive, ostensive.
Five rules of definition	Essential attributes; not circular; not too broad/narrow; clear language; affirmative where possible.

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